COMPETENCY STANDARD

FOR

ELECTRICAL WORKS IN ENERGY SECTOR

(ENERGY SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the effective performance of an occupation or trade role, corresponding to the standards of performance required in the workplace.

A Competency Standard:

- Provides a consistent and reliable set of components for training, recognizing, and assessing workers' skills
- Enables industry-recognized qualifications to be awarded through direct assessment of practical workplace competencies
- Encourages the development and delivery of flexible training that suits individual needs and industry requirements
- Supports learning and assessment in a work-related environment leading to verifiable workplace outcomes

Competency Standards describe the skills, knowledge, and attitudes needed to perform effectively in the workplace. They acknowledge that workers can achieve vocational and technical competency through many pathways, emphasizing what the learner can DO rather than how or where they learned it.

This Competency Standard for Electrical Works in the Energy Sector has been developed with the following key considerations:

- Educational Background: Maximum Grade SSC pass.
- Industry Context: Bangladesh's energy sector including power plants, substations, distribution networks, gas processing facilities, and renewable energy sites.
- Language of Delivery: Training may be delivered in Bangla; oral and practical assessment methods are prioritized

A Unit of Competency describes a distinct, work-related activity normally undertaken by one person in accordance with industry standards. Each unit is documented in a standard format comprising:

- Unit Code, Title, and Descriptor
- Elements of Competency and Performance Criteria
- Range of Variables
- Curricular Content Guide
- Assessment Evidence Guide

Units & Elements at a Glance:

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-BPI-EWE-01-G	Apply occupational health and safety (ohs) practices at energy worksites	1. Identify OHS policies and procedures. 2. Use personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-BPI-EWE-02-G	Carry Out Workplace Interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	10
SICIP-BPI-EWE-03-G	Operate In a Team Environment	1 Identify own role and responsibilities within the team 2 Support and cooperate with team members 3 Communicate and share information within the team 4 Contribute to problem-solving within the team	10
		Total Hour	**30 Hrs.**

Sector Specific Competencies (10 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-BPI-EWE-01-S	Use hand tools and basic power tools for electrical work	1 Identify and inspect tools before use 2 Use hand tools safely and correctly 3 Operate basic power tools safely under supervision 4 Clean and store tools properly after use	10
		Total Hour	**10 Hrs.**

Occupation Specific Competencies (320 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-BPI-EWE-01-O	Identify Electrical Systems and Components in the Energy Sector	1. Identify basic Bangladesh electrical safety rules and regulations 2. Identify electrical materials, cables, and accessories 3. Identify basic electrical tools and testing instruments 4. Read and interpret simple electrical diagrams 5. Identify key electrical equipment at energy sector facilities.	35
SICIP-BPI-EWE-02-O	Implement Electrical Safety Practices in Energy Worksites	1. Apply electrical safety procedures before starting work 2. Identify electrically hazardous areas and conditions 3. Follow lockout/tagout (LOTO) procedures 4. Comply with national electrical safety regulations and codes	30
SICIP-BPI-EWE-03-O	Implement Basic Electrical Circuits and Electrical Wiring System	1. Interpret electrical symbols, drawings and manual 2. Conduct DC circuit experiments 3. Construct and implement AC circuits 4. Prepare materials and wiring routes 5. Install cables, conduits and trunking circuits 6. Connect switches, sockets and lighting circuits 7. Inspect completed wiring installations	45
SICIP-BPI-EWE-04-O	Install Electrical Systems in Energy Worksites	1. Install cables, cable trays, LT/HT meters 2. Install power distribution components 3. Install distribution boards 4. Terminate and connect power and control cables 5. Install power supply	60

Code	Unit of Competency	Elements of Competency	Duration (hours)
		components 6. Carry out wiring and testing 7. Inspect installation quality and alignment with design	
SICIP-BPI-EWE-05-O	Operate and Maintain Electrical Equipment	1. Understand working principles of generator and transformer 2. Operate generator and associated control systems 3. Operate transformer and associated switchgear 4. Understand Battery Bank and battery charger. 5. Perform routine maintenance of electrical equipment 6. Diagnose faults and carry out basic troubleshooting	80
SICIP-BPI-EWE-06-O	Install Earthing System	1. Interpret earthing system requirements 2. Prepare tools, equipment, and materials 3. Prepare site for earthing installation 4. Install earthing electrode and components 5. Test earthing system performance 6. Complete installation and documentation	30
SICIP-BPI-EWE-07-O	Perform Motor Connection	1. Identify and select control and protective devices 2. Collect tools, equipment and materials 3. Install control and protective devices 4. Perform motor connection 5. Check and test circuit 6. Clean the workplace	40
		Total Hours	**0 Hrs.**

COMPETENCY STANDARD: ELECTRICAL WORKS IN ENERGY SECTOR

Generic Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES AT ENERGY WORKSITES**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-BPI-EWE-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Apply occupational health and safety (OHS) practices at energy worksites. It specifically includes the tasks of identifying OHS policies and procedures, using personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **OHS policies** and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Use personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including **personal protective equipment (PPE).** 2.2 Common health issues are recognized. 2.3 Common safety issues are identified and applied.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted 3.3 Hazards and risk assessment are reported
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 **Emergency response plans and procedures** are responded to. 4.3 **First aid procedures** during emergency situations are identified.

Range of variables:

Variables	**Range** (may include but not limited to)
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective equipment	2.1 Safety glasses 2.2 Ear plugs 2.3 Gloves 2.4 Apron

	2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, signs and symbols 4.3 Codes of conduct 4.4 Projector 4.5 Learning manual 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT WORKPLACE INTERACTION**	Nominal Duration: 10 hrs.	Unit Code: SICIP-BPI-EWE-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **courteous manner** to gather and convey information. 1.4 **Workplace procedures and matters** are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from **appropriate sources**. 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette

2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	3.1 Human Resources (HR) Department 3.2 Managers 3.3 Supervisors 3.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	2.1 Demonstrating workplace communication and etiquette 2.2 Interpreting workplace instructions and symbols 2.3 Demonstrating active participation in workplace meeting 2.4 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-BPI-EWE-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying own role and responsibilities within the team, supporting and cooperating with team members, communicating and sharing information within the team and contributing to problem-solving within the team.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify own role and responsibilities within team	1.1. Personal role and responsibilities are identified within the team environment. 1.2. Importance of relationships within and outside the team are interpreted . 1.3. Roles and objectives of the team are identified and interpreted. 1.4. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Support and cooperate with team members	2.1. **<u>Team members</u>** are supported by sharing information, resources, and assistance as needed. 2.2. Tasks are carried out cooperatively to achieve common work objectives. 2.3. Clear and respectful communication is maintained with all team members. 2.4. Individual responsibilities are fulfilled while contributing to team outcomes. 2.5. Differences are handled constructively to maintain a positive working environment. 2.6. Workplace rules and team procedures are followed at all times.
3. Communicate and share information within the team	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through **<u>sharing information</u>** or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Contribute to problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are

	recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Team members	1.1 Supervisors and instructors 1.2 Skilled technicians and electricians 1.3 Co-workers and trainees 1.4 Support staff and helpers 1.5 Representatives from other departments
2 Sharing information	2.1 Agenda 2.2 Minutes 2.3 Progress and incident reports 2.4 Operational manuals 2.5 Visual and graphic materials 2.6 Emails and SMS 2.7 Policy, procedure and standards 2.8 OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	2.1. Identifying own role and responsibilities within team 2.2. Communicating and co-operating with team members 2.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Workplace documents, Projector 4.3. Learning manual 4.4. Tools, equipment and facilities appropriate to the process or activities. 4.5. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **USE HAND TOOLS AND BASIC POWER TOOLS FOR ELECTRICAL WORK**	Nominal Duration: 10 hrs.	Unit Code: SICIP-BPI-EWE-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand tools and basic power tools for electrical work. It specifically includes the task of identifying and inspecting tools, using hand tools before use, using hand tools safely and correctly, operating basic power tools safely under supervision and cleaning and storing tools properly after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect tools before use	1.1 Appropriate **<u>hand tools</u>** and **<u>power tools</u>** are identified based on the task requirements. 1.2 Tools are inspected for physical damage, wear, or defects before use. 1.3 Electrical safety of tools is checked, including insulation, plugs, and cables. 1.4 Correct tool type and size are selected to ensure safe and efficient operation. 1.5 **<u>Faulty or unsafe tools</u>** are reported and removed from use. 1.6 Tools are confirmed to be clean, functional, and ready for operation.
2. Use hand tools safely and correctly	2.1. Hand tools are used according to their intended purpose and manufacturer guidelines. 2.2. **<u>Correct techniques</u>** are applied to ensure safe and efficient operation. 2.3. Proper grip, posture, and control are maintained during use. 2.4. **<u>Personal protective equipment</u>** is used as required. 2.5. Excessive force and unsafe practices are avoided to prevent injury or damage. 2.6. Tools are handled carefully to ensure safety of self and others.
3. Operate basic power tools under supervision	3.1. Basic power tools are operated under supervision following instructions and safety guidelines. 3.2. Tools are checked and set up correctly before operation. 3.3. **<u>Safe handling and correct operating techniques</u>** are applied during use. 3.4. Work is carried out with control to avoid accidents or damage. 3.5. Any issues or unsafe conditions are reported immediately.

4. Clean and store tools properly after use	4.1. Tools are cleaned after use to remove dust, debris, and residues. 4.2. Tools are inspected for damage or wear before storage. 4.3. **Faulty or damaged tools** are reported and separated from usable tools. 4.4. Tools are stored in designated locations following workplace procedures. 4.5. Storage is arranged to ensure safety, accessibility, and prevention of damage. 4.6. Work area is left clean and organized after completion.

Range of Variables

Variable	**Range** May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Faulty or unsafe tools	3.1 Broken or cracked handles 3.2 Damaged insulation or exposed conductors 3.3 Loose parts or malfunctioning switches 3.4 Overheating or unusual noise in power tools 3.5 Worn-out or ineffective cutting edges
4. Correct techniques	4.1 Applying appropriate force and direction 4.2 Using the correct angle and positioning 4.3 Selecting proper tool size for the task

	4.4 Maintaining control during cutting, tightening, or loosening 4.5 Using both hands where required for stability
5. Personal protective equipment	5.1 Safety gloves (insulated or cut-resistant) 5.2 Safety glasses or face shield 5.3 Safety shoes 5.4 Protective clothing 5.5 Hearing protection (if required)
6. Safe handling and correct operating techniques	6.1 Holding tools firmly with correct grip 6.2 Maintaining stable posture and balance 6.3 Operating at appropriate speed and pressure 6.4 Keeping hands clear of moving parts 6.5 Switching off before adjustments or changes
7. Faulty or damaged tools	7.1 Broken handles or loose components 7.2 Worn-out or blunt cutting edges 7.3 Damaged insulation or exposed wires 7.4 Malfunctioning switches or motors 7.5 Excessive vibration or unusual noise

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concepts of hand and power tools and their uses. 1.2 Procedures for checking tools for damage and safety before use. 1.3 Identify correct tool type, size, and condition. 1.4 Procedures for proper grip, posture, and technique. 1.5 Identify required PPE and safe work practices. 1.6 Understand risks of incorrect use and excessive force. 1.7 Identify faults and unsafe conditions during use. 1.8 Understand importance of control and reporting issues. 1.9 Identify damaged tools and correct storage locations. 1.10 Understand importance of cleanliness, safety, and organization.
2. Underpinning Skills	2.1 Applying correct tool selection based on task requirements to ensure safety and efficiency 2.2 Performing electrical safety checks on tools, including cables, plugs, and insulation 2.3 Using only clean, functional, and serviceable tools while reporting faulty ones 2.4 Applying proper handling techniques, grip, posture, and control during operation 2.5 Operating basic power tools under supervision following safety procedures 2.6 Performing tasks in a controlled and safe manner to prevent accidents and damage 2.7 Maintaining tools by cleaning, inspecting, and storing them properly after use
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and inspected tools before use 1.2 used hand tools safely and correctly 1.3 operated basic power tools under supervision 1.4 cleaned and stored tools properly after use
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency:	Nominal Duration:	Unit Code:
IDENTIFY ELECTRICAL SYSTEMS AND COMPONENTS IN THE ENERGY SECTOR	35 Hrs.	SICIP-BPI-EWE-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Identify electrical systems and components in the energy sector. It specifically includes the task of identifying basic Bangladesh electrical safety rules and regulations, identifying electrical materials, cables, and accessories, identifying basic electrical tools and testing instruments, reading and interpreting simple electrical diagrams and identifying key electrical equipment at energy sector facilities.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify basic Bangladesh electrical safety rules and regulations	1.1 Key national electrical safety rules applicable to energy sector work in Bangladesh are identified with guidance. 1.2 **<u>Bangladesh Electricity Rules and regulation</u>** on safe voltages, earthing, and insulation requirements are recognized. 1.3 Electrical safety warning signs and labels used at power facilities are identified and interpreted correctly. 1.4 Safety requirements for working near live electrical systems are described and followed.
2. Identify electrical materials, cables, and accessories	2.1 **<u>Common electrical materials</u>** used in energy sector installation works are identified. 2.2 **<u>Different types of cables</u>** are identified and distinguished. 2.3 **<u>Major electrical accessories</u>** are identified and sorted correctly. 2.4 Appropriate cable for the task is selected based on supervisor guidance and job specifications.
3. Identify basic electrical tools and testing instruments	3.1 **<u>Common electrical hand tools</u>** are identified. 3.2 **<u>Basic testing instruments</u>** are identified. 3.3 Testing instruments are handled and stored carefully without causing damage. 3.4 The supervisor is consulted before using any unfamiliar tool or instrument.
4. Read and interpret simple electrical diagrams with guidance	4.1 Common electrical symbols used in wiring diagrams and single-line diagrams are identified. 4.2 The overall layout and main connections shown in a simple circuit diagram are described. 4.3 Equipment locations shown on a basic layout drawing are identified and matched to physical items in the facility. 4.4 Any confusion about diagram interpretation is reported to

	the supervisor for clarification.
5. Identify key electrical equipment at energy sector facilities	5.1 **<u>Main electrical equipment</u>** at energy facilities is identified. 5.2 Basic functions of each equipment item are described. 5.3 Equipment nameplates and rating labels are read to identify voltage, current, and power ratings with supervisor assistance. 5.4 Battery banks, UPS units, and solar panels used as backup power at energy facilities are identified and their purpose is described.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Bangladesh Electricity Rules and regulation	1.1 Bangladesh Electricity Rules, 1937 (amended) 1.2 Bangladesh National Building Code (BNBC) electrical provisions 1.3 Bangladesh Power Development Board (BPDB) safety requirements 1.4 Electrical earthing and grounding rules 1.5 Fire safety regulations for electrical systems
2. Common electrical materials	2.1 Insulation tape 2.2 Cable glands 2.3 Cable lugs 2.4 Ferrules 2.5 Conduits (PVC/GI) 2.6 Flexible conduit 2.7 Cable trays and supports 2.8 PVC insulated wire (1.0–6 sqmm) 2.9 Flexible wire 2.10 Armoured cable 2.11 Conduit fittings (bend, tee) 2.12 Earthing rods 2.13 Earthing wire/strip 2.14 Explosion-proof fittings 2.15 Wire markers and labels 2.16 Cable ties 2.17 Gaskets and sealing compounds
3. Different types of cables	3.1 XLPE insulated cables 3.2 PVC insulated cables 3.3 Flexible cables 3.4 Fire-resistant cables 3.5 Shielded cables
4. Major electrical accessories	4.1 Switch (1-way, 2-way, intermediate) 4.2 Socket outlet (2-pin, 3-pin) 4.3 Ceiling rose 4.4 Lamp holder (batten, pendant) 4.5 Fan regulator

	4.6 Bell push switch 4.7 Dimmer switch 4.8 Junction box 4.9 Switch box (PVC/metal) 4.10 Gang box 4.11 Distribution board accessories (cover, door) 4.12 Terminal block 4.13 Neutral link 4.14 Earth bar 4.15 Indicator light 4.16 Push button switch 4.17 Selector switch 4.18 MCB, MCCB
5. Common electrical hand tools	5.1 Screwdrivers (insulated) 5.2 Combination pliers 5.3 Wire stripper and cutter 5.4 Crimping pliers 5.5 Tester 5.6 Voltage tester 5.7 Cable pulling tools 5.8 Drilling machine 5.9 Hack Saw 5.10 Pipe cutter
6. Basic testing instruments	6.1 Multimeter 6.2 Clamp meter 6.3 Watt meter 6.4 Insulation resistance tester (Megger) 6.5 Earth resistance tester 6.6 Phase sequence tester 6.7 Continuity tester 6.8 OHM tester 6.9 Voltage detector 6.10 Power quality analyzer 6.11 Loop impedance tester
7. Main electrical equipment	7.1 Generators 7.2 Transformers 7.3 Distribution boards 7.4 Circuit breakers 7.5 Switchgear 7.5.1 CB (Circuit Breaker). 7.5.2 LBS (Load Break Switch). 7.5.3 Relays. 7.5.4 Fuses. 7.5.5 CT (Current Transformer). 7.5.6 PT (Potential Transformer).

Curricular Content Guide

1. Underpinning Knowledge	1.1. Concepts of basic electrical safety rules and regulations in Bangladesh. 1.2. Procedures for following safety requirements near electrical systems. 1.3. Identify types of cables and common electrical accessories. 1.4. Understand correct selection of materials for safe installation. 1.5. Procedures for safe handling and storage of tools and instruments. 1.6. Understand importance of consulting supervisor for unfamiliar equipment. 1.7. Procedures for reading and interpreting basic diagrams. 1.8. Procedures for identifying equipment and reading nameplates. 1.9. Understand basic functions and purpose of electrical equipment.
2. Underpinning Skills	2.1 Applying basic Bangladesh electrical safety rules and regulations in energy sector work 2.2 Following safe practices for working near live electrical systems and recognizing warning signs 2.3 Selecting appropriate cables based on task requirements and supervisor guidance 2.4 Using basic electrical tools and testing instruments safely and correctly 2.5 Following proper handling and storage procedures for tools and instruments 2.6 Reporting uncertainties and seeking supervisor support when required 2.7 Identifying key electrical equipment and explaining their basic functions and ratings 2.8 Recognizing backup power systems such as battery banks, UPS, and solar panels.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified basic Bangladesh electrical safety rules and regulations 1.2 identified electrical materials, cables, and accessories 1.3 identified basic electrical tools and testing instruments 1.4 read and interpreted simple electrical diagrams 1.5 identified key electrical equipment at energy sector facilities.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IMPLEMENT ELECTRICAL SAFETY PRACTICES IN ENERGY WORKSITES**	Nominal Duration: 30 Hrs.	Unit Code: SICIP-BPI-EWE-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Implement electrical safety practices at energy sector worksites. It specifically includes the task of applying electrical safety procedures before starting work, identify electrically hazardous areas and conditions, following lockout/tagout (LOTO) procedures and complying with national electrical safety regulations and codes.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply electrical safety procedures before starting work	1.1 **Reasons for electrical hazards** in the work area are identified. 1.2 Required PPE is selected, checked, and correctly worn before approaching electrical equipment or cables. 1.3 Confirmation is obtained from the supervisor that power is switched off and isolated before touching any electrical component. 1.4 Barriers, warning signs, and safety tape are placed correctly around the work area as instructed by the supervisor.
2. Identify electrically hazardous areas and conditions	2.1 Electrical work areas are inspected to identify potential hazards. 2.2 **Environmental risks** are recognized and assessed. 2.3 **Warning signs, labels, and safety markings** are checked and interpreted to determine hazard levels. 2.4 **Unsafe conditions** are reported promptly in accordance with workplace safety procedures. 2.5 Risk levels are evaluated following electrical safety standards. 2.6 Work is planned considering identified hazards to prevent electric shock, fire, and equipment damage
3. Follow lockout/tagout (LOTO) procedures	3.1 The purpose and importance of **lockout/tagout (LOTO) procedures** are explained in simple terms. 3.2 Correct isolation of a circuit or equipment is confirmed by checking that power is off using a voltage tester before work begins. 3.3 Lockout/tagout tags and locks are applied to isolated equipment as instructed by the supervisor.
4. Comply with national electrical safety regulations and codes	4.1 **Work activities** are carried out in accordance with applicable regulations and approved codes of practice. 4.2 Required permits, safety clearances, and work authorizations are confirmed with the supervisor before starting any electrical task.

	4.3 **<u>Corrective actions</u>** are taken within the limits of responsibility to maintain compliance with safety regulations and codes. 4.4 Any breach of safety rules noticed at the worksite is reported immediately to the supervisor. 4.5 Records and documentation are completed and maintained according to organizational requirements.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Reasons for electrical hazards	1.1 Poor insulation 1.2 Faulty wiring 1.3 Overloaded circuits 1.4 Improper grounding 1.5 Damaged electrical equipment 1.6 Wet or damp working conditions 1.7 Lack of maintenance 1.8 Use of substandard materials 1.9 Loose connections 1.10 Exposed live parts 1.11 Improper use of electrical tools 1.12 Lack of training or awareness 1.13 Absence of protective devices (fuse, MCB, ELCB) 1.14 Poor installation practices 1.15 Unauthorized or unsafe modifications
2. Environmental risks	2.1 High humidity or rain exposure 2.2 Excessive heat or temperature fluctuations 2.3 Dust, fumes, or corrosive substances 2.4 Poor ventilation 2.5 Confined spaces 2.6 Presence of flammable materials
3. Warning signs, labels, and safety markings	3.1 High voltage warning signs 3.2 Danger and caution labels 3.3 Equipment rating plates 3.4 Lockout/Tagout (LOTO) tags 3.5 Color-coded safety markings (e.g., red for danger, yellow for caution) 3.6 Electrical hazard symbols
4. Unsafe conditions	4.1 Unsecured electrical panels 4.2 Missing protective covers 4.3 Improper use of extension cords 4.4 Lack of personal protective equipment (PPE) 4.5 Unauthorized access to restricted areas 4.6 Improper grounding practices
5. Lockout/Tagout (LOTO) procedures	5.1 Identify energy sources 5.2 Shut down equipment 5.3 Isolate energy

	5.4 Release stored energy 5.5 Verify isolation 5.6 Restore and re-energize
6. Work activities	6.1 Installation of wiring systems and electrical components 6.2 Maintenance and servicing of electrical equipment 6.3 Testing and commissioning of systems 6.4 Troubleshooting and fault rectification 6.5 Inspection of electrical installations
7. Corrective actions	7.1 Isolating unsafe equipment 7.2 Replacing damaged cables or components 7.3 Adjusting improper connections 7.4 Using appropriate PPE and tools 7.5 Stopping work in case of unsafe conditions 7.6 Reporting issues beyond authority to higher personnel

Curricular Content Guide

1. Underpinning Knowledge	1.1. Concepts of electrical hazards and safety procedures before starting work. 1.2. Identify hazards, warning signs, and unsafe conditions. 1.3. Understand importance of safety preparation to prevent accidents. 1.4. Identify risk levels, safety markings, and dangerous conditions. 1.5. Understand need for hazard reporting and safe work planning. 1.6. Procedures for isolating power and applying LOTO devices. 1.7. Identify correct use of tags, locks, and voltage testers. 1.8. Understand importance of preventing accidental energizing. 1.9. Identify compliance requirements and safety breaches. 1.10. Understand importance of adhering to regulations and reporting issues.
2. Underpinning Skills	2.1. Applying electrical safety procedures before starting work by identifying hazards and isolating power 2.2. Following safe work practices near electrical equipment, including barriers and warning signs 2.3. Interpreting safety signs, labels, and markings to assess risk levels 2.4. Reporting unsafe conditions promptly according to workplace procedures 2.5. Performing work planning based on identified hazards to prevent accidents and damage 2.6. Following lockout/tagout (LOTO) procedures to ensure proper isolation of electrical systems 2.7. Using voltage testers to confirm power is off before starting work 2.8. Reporting safety breaches and maintaining required

	safety documentation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied electrical safety procedures before starting work 1.2 identified electrically hazardous areas and conditions 1.3 followed lockout/tagout (LOTO) procedures 1.4 complied with national electrical safety regulations and codes
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IMPLEMENT BASIC ELECTRICAL CIRCUITS AND ELECTRICAL WIRING SYSTEM**	Nominal Duration: 45 Hrs.	Unit Code: SICIP-BPI-EWE-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Implement basic electrical circuits and electrical wiring system. It specifically includes the task of interpreting electrical symbols, drawings and manual, conducting DC circuit experiments, constructing and implementing AC circuits, preparing materials and wiring routes, installing cables, conduits and trunking circuits, connecting switches, sockets and lighting circuits and inspecting completed wiring installations.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret electrical symbols, drawings and manual	1.1 **Electrical symbols** used in circuit diagrams are identified and understood. 1.2 **Electrical drawings** are identified to implement wiring and troubleshooting procedures. 1.3 Electrical manuals are identified and understood. 1.4 Wiring diagrams and schematics are understood. 1.5 Common electrical standards and codes are recognized.
2. Conduct DC circuit experiments	2.1 Ohm's Law is verified by measuring voltage, current, and resistance in a DC circuit. 2.2 Series circuits are constructed and tested. 2.3 Parallel circuits are constructed and tested. 2.4 Combined circuits (series and parallel) are created. 2.5 **Circuit theorems** are implemented and verified.
3. Construct and implement AC circuits	3.4 RL circuits are constructed and implemented. 3.5 RC circuits are constructed and implemented. 3.6 Resistor (R) and capacitor (C) components are connected. 3.7 RLC circuits are constructed and implemented. 3.8 **Resonance circuits** are designed and implemented. 3.9 **Filter circuits** are constructed and tested.
4. Prepare materials and wiring routes	4.1 Required **materials, tools, and accessories** are identified based on wiring needs and circuit design. 4.2 Materials are checked to ensure they are in good condition and meet required standards. 4.3 **Wiring routes** are planned according to layout, safety rules, and site conditions. 4.4 Measurements and markings are made accurately for cables, switches, and fittings. 4.5 Suitable paths for conduits or casing are selected for safe and neat installation.
5. Install cables, conduits	5.1 **Cables, conduits, and trunking** are installed according

and trunking circuits	to wiring diagrams and specifications. 5.2 Conduits and trunking are cut, fitted, and secured properly to ensure durability and alignment. 5.3 Cables are pulled and laid neatly without damage, twisting, or excessive bending. 5.4 **Connections and joints** are made securely following standard procedures and safety practices. 5.5 Proper spacing, support, and routing are maintained for safe and organized installation. 5.6 Installation is checked to ensure compliance with electrical standards and quality requirements.
6. Connect switches, sockets and lighting circuits	6.1 **Switches, sockets, and lighting components** are installed in correct positions as per layout and specifications. 6.2 **Wiring connections** are made accurately following circuit diagrams and standard color codes. 6.3 Terminals are tightened securely to ensure proper electrical contact and safety. 6.4 Connections are checked to avoid loose wiring, short circuits, or incorrect polarity. 6.5 All fittings are aligned, fixed properly, and finished neatly. 6.6 Work is carried out in compliance with safety rules and electrical standards.
7. Inspect completed wiring installations	7.1 Completed wiring installation is visually inspected for correctness, neatness, and compliance with layout. 7.2 **Connections and terminations** are checked to ensure they are secure and properly fitted. 7.3 Continuity, insulation, and polarity are verified using appropriate **testing tools**. 7.4 Any faults, defects, or deviations are identified and corrected. 7.5 Installation is confirmed to meet safety standards and operational requirements. 7.6 Work area is cleaned and documentation is completed as required.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Electrical symbols	1.1 Source (AC/DC) 1.2 Resistor 1.3 Capacitor 1.4 Inductor 1.5 Switches 1.6 Relay coils 1.7 Magnetic contactor 1.8 Circuit breaker 1.9 Timer/counter

	1.10 Overload
2. Electrical drawings	2.1 Layout diagram 2.2 Schematic diagram 2.3 Wiring diagram
3. Circuit theorems	3.1 Kirchhoff's voltage law 3.2 Kirchhoff's current law 3.3 Thevenin's theorem 3.4 Superposition
4. Resonance circuits	4.1 Series resonance 4.2 Parallel resonance
5. Filter circuits	5.1 Active filter 5.2 Passive filter
6. Materials, tools, and accessories	6.1 **Materials** 6.1.1 Electrical cables and wires (PVC, XLPE, flexible wires) 6.1.2 Conduits (PVC, metal) 6.1.3 Casing and capping 6.2 **Accessories** 6.2.1 Switches 6.2.2 Sockets 6.2.3 Plugs 6.2.4 Junction boxes 6.2.5 Distribution boards 6.2.6 MCBs 6.2.7 Fuses 6.2.8 Cable clips 6.2.9 Saddles 6.2.10 Connectors 6.2.11 Lugs 6.3 **Tools** 6.3.1 Pliers 6.3.2 Screwdrivers 6.3.3 Wire strippers 6.3.4 Testers 6.3.5 Measuring tape 6.3.6 Drilling machine
7. Wiring routes	7.1 Surface wiring or concealed wiring 7.2 Horizontal and vertical routing alignment 7.3 Avoidance of heat, moisture, and mechanical damage areas 7.4 Accessibility for maintenance and inspection 7.5 Separation from water lines and other services
8. Cables, conduits, and trunking	8.1 Electrical cables (PVC insulated, XLPE, flexible cables) 8.2 Single-core and multi-core cables 8.3 PVC conduits (rigid and flexible) 8.4 Metal conduits (GI, steel) 8.5 Trunking systems (PVC trunking, metal trunking) 8.6 Cable trays and ducts (where applicable)

9. Connections and joints	9.1 Straight joints and tee joints 9.2 Terminal connections at switches, sockets, and DBs 9.3 Crimping lugs and ferrules 9.4 Soldered or mechanical connections 9.5 Use of junction boxes for jointing
10. Switches, sockets, and lighting components	10.1 One-way, two-way, and intermediate switches 10.2 Socket outlets (2-pin, 3-pin, industrial sockets) 10.3 Lamp holders (batten, pendant, angle holder) 10.4 LED lights, tube lights, ceiling lights 10.5 Distribution boards and control devices
11. Wiring connections	11.1 Phase (live), neutral, and earth connections 11.2 Series and parallel circuit connections 11.3 Loop-in wiring system 11.4 Connections at switches, sockets, and lighting points 11.5 Termination inside junction boxes and distribution boards
12. Connections and terminations	12.1 Tightness of terminals at switches, sockets, and DBs 12.2 Proper use of lugs, ferrules, and connectors 12.3 Correct phase, neutral, and earth connections 12.4 Secure fixing junction boxes and panels 12.5 No exposed conductors or loose ends
13. Testing tools	13.1 Multimeter 13.2 Insulation resistance tester (Megger) 13.3 Continuity tester or test lamp 13.4 Voltage tester

Curricular Content Guide

1. Underpinning Knowledge	1.1. Procedures for reading wiring diagrams and schematics. 1.2. Identify symbols, drawings, and manuals used in electrical work. 1.3. Identify voltage, current, resistance, and circuit components. 1.4. Understand application of Ohm's Law and circuit behavior. 1.5. Procedures for constructing and testing AC and resonance circuits. 1.6. Identify resistors, capacitors, and inductors. 1.7. Understand circuit response, resonance, and filtering. 1.8. Identify correct methods for cable laying and jointing. 1.9. Understand need for safe, neat, and standard-compliant installation. 1.10. Procedures for correct installation and wiring connections.
2. Underpinning Skills	2.1 Applying basic electrical principles by performing DC circuit experiments and verifying results 2.2 Constructing and testing series, parallel, and combined DC circuits 2.3 Performing AC circuit construction including RL, RC, and RLC circuits and resonance applications 2.4 Using electrical components such as resistors and

	capacitors correctly in circuit design 2.5 Planning wiring routes and performing accurate measurement and marking for installation 2.6 Installing cables, conduits, and trunking systems following standards and specifications 2.7 Performing safe and correct connections of switches, sockets, and lighting circuits 2.8 Inspecting wiring installations through visual checks and electrical testing procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted electrical symbols, drawings and manual 1.2 conducted DC circuit experiments 1.3 constructed and implemented AC circuits 1.4 prepared materials and wiring routes 1.5 installed cables, conduits and trunking circuits 1.6 connected switches, sockets and lighting circuits 1.7 inspected completed wiring installations
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INSTALL ELECTRICAL SYSTEMS IN ENERGY WORKSITES**	**Nominal Duration:** 60 Hrs.	**Unit Code:** SICIP-BPI-EWE-04-O
Unit Descriptor:		

This unit covers the skills, knowledge and attitudes required to install electrical systems in energy worksites. It specifically includes the task of installing cables, cable trays, LT/HT meters, installing power distribution components, installing distribution boards, terminating and connect power and control cables, installing power supply components, carried out wiring and testing and inspecting installation quality and alignment with design.

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install cables, cable trays, LT/HT meters	1.1 **Work instructions, drawings, and specifications** are interpreted. 1.2 **Cables, cable trays, and LT/HT meters** are identified and selected. 1.3 Cable trays and supports are installed according to layout and load requirements. 1.4 Cables are laid, dressed, and terminated using appropriate **tools** and methods. 1.5 LT/HT meters are installed, connected, and calibrated in accordance with electrical standards. 1.6 Earthing and bonding are completed to ensure electrical safety and system reliability. 1.7 Installation work is checked and tested.
2. Install power distribution components	2.1 Work instructions, single-line diagrams, and layout drawings are interpreted. 2.2 Distribution boards, switchgear, circuit breakers, isolators, and protective devices are selected. 2.3 **Mounting locations** are prepared and components are installed and fixed as per design and load considerations. 2.4 **Electrical connections** are completed using approved wiring methods. 2.5 Protection systems are configured to meet load and safety requirements. 2.6 Earthing and bonding of distribution components are carried out to ensure safe operation. 2.7 Installed components are tested, verified, and documented.
3. Install distribution boards	3.1 Drawings and specifications are interpreted to determine location, type, and rating of distribution boards. 3.2 Distribution boards are positioned, aligned, and securely mounted following installation standards. 3.3 **Incoming and outgoing circuits** are connected ensuring correct phase, neutral, and earthing arrangements. 3.4 **Protective devices** are installed and configured according to load and safety requirements. 3.5 Installation is tested, inspected, and verified for compliance with electrical and safety standards.

4. Terminate and connect power and control cables	4.1 Cable schedules, wiring diagrams, and termination details are interpreted. 4.2 Cables are prepared, stripped, and fitted with appropriate lugs, glands, and ferrules using proper tools. 4.3 Power and control cables are terminated and connected. 4.4 Connections are tightened, insulated, and secured. 4.5 Continuity, insulation resistance, and functional checks are carried out.
5. Install power supply components	5.1 Work instructions and technical drawings are interpreted. 5.2 **Components** are selected and inspected for compliance with standards. 5.3 Components are positioned, aligned, and securely mounted according to layout. 5.4 **Electrical connections** are completed ensuring correct input-output configuration, polarity, and earthing. 5.5 Installed components are tested and verified to ensure proper operation with safety regulations.
6. Carry out wiring and testing	6.1 Wiring diagrams and circuit layouts are interpreted to determine routing and connection requirements. 6.2 **Wiring** is carried out using approved methods. 6.3 Connections are completed with proper identification, tight terminations, and compliance with standards. 6.4 Electrical tests are performed using appropriate instruments. 6.5 Test results are recorded and any faults identified are rectified to ensure safe and compliant system operation.
7. Inspect installation quality and alignment with design	7.1 Completed installation is visually inspected against drawings and specifications. 7.2 Alignment, positioning, and mechanical fixing of components are checked. 7.3 Electrical connections, labeling, and routing are verified for correctness, safety, and neatness. 7.4 Testing and measurement results are reviewed to confirm system performance meets specified parameters. 7.5 **Non-conformities** are identified, reported, and corrective actions are taken to achieve compliance.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Work instructions, drawings, and specifications	1.1 Electrical layout drawings and cable routing plans 1.2 Single-line diagrams (SLD) 1.3 Schematic diagrams and wiring details 1.4 Technical specifications and BOQ 1.5 Manufacturer manuals and installation guidelines

	1.6 Method statements and work procedures
2. Cables, cable trays, and LT/HT meters	2.1 **Power cables** 2.1.1 PVC insulated cables 2.1.2 XLPE insulated cables 2.1.3 Armoured cables 2.1.4 Unarmoured cables 2.2 **Control and instrumentation cables** 2.2.1 Control cables (Single transmission) 2.2.2 Instrumentation cables (Measurement systems) 2.3 **Cable trays** 2.3.1 Ladder type cable tray 2.3.2 Perforated cables tray 2.3.3 Solid bottom cable tray 2.4 **Cable supports** 2.4.1 Brackets 2.4.2 Hangers 2.4.3 Clamps 2.5 LT (Low Tension) meters for low voltage systems 2.6 HT (High Tension) meters for high voltage systems
3. Tools	3.1 Cable pulling tools 3.1.1 Rollers 3.1.2 Winches 3.1.3 Fish tape 3.2 Crimping and stripping tools 3.3 Measuring and marking tools 3.4 Testing instruments 3.4.1 Multimeter 3.4.2 Insulation tester
4. Mounting locations	4.1 Wall-mounted or floor-mounted panels 4.2 Indoor or outdoor enclosures 4.3 Adequate clearance for operation and maintenance 4.4 Proper alignment and leveling 4.5 Secure fixing using anchors, bolts, and brackets 4.6 Accessibility and ventilation considerations
5. Electrical connections	5.1 Busbar connections and cable terminations 5.2 Use of lugs, ferrules, and glands 5.3 Proper phase, neutral, and earth connections 5.4 Tightening torque as per manufacturer recommendations
6. Incoming and outgoing circuits	6.1 Incoming supply from main source or feeder 6.2 Outgoing circuits to lighting, sockets, or equipment 6.3 Phase (L), neutral (N), and earth (E) connections 6.4 Busbar connections (phase, neutral, earth busbars) 6.5 Use of cable lugs, ferrules, and glands 6.6 Proper cable identification and labeling
7. Protective devices	7.1 Miniature Circuit Breakers (MCB) 7.2 Moulded Case Circuit Breakers (MCCB) 7.3 Residual Current Circuit Breakers (RCCB) 7.4 Residual Current Breaker with Overcurrent (RCBO)

	7.5 Fuses and surge protection devices (SPD)
8. Components	8.1 Control devices 8.1.1 Contactors 8.1.2 Relays 8.1.3 Timers 8.2 Protection devices 8.2.1 MCB 8.2.2 MCCB 8.2.3 RCCB 8.2.4 Fuse 8.3 Switches 8.4 Push buttons 8.5 Selector switches 8.6 Indicators 8.7 Pilot lamps 8.8 Alarms 8.9 Terminal blocks and connectors 8.10 Sensors and basic automation devices
9. Electrical connections	9.1 Power and control circuit connections 9.2 Input-output (I/O) terminal connections 9.3 Phase, neutral, and earth connections 9.4 Use of lugs, ferrules, and terminal blocks 9.5 Wiring according to diagrams and labeling standards
10. Wiring	10.1 Surface wiring and concealed wiring 10.2 Conduit wiring 10.3 Trunking and casing systems 10.4 Cable tray and duct wiring 10.5 Panel internal wiring using ducts and terminal blocks
11. Non-conformities	11.1 Deviation from drawings or specifications 11.2 Loose or incorrect connections 11.3 Improper labeling or identification 11.4 Mechanical misalignment or poor fixing 11.5 Test results outside acceptable limits

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concepts of cables, cable trays, LT/HT meters, and earthing systems. 1.2 Procedures for installing, terminating, and testing cables, meters, and wiring systems. 1.3 Identify components, tools, switchgear, circuit breakers, and protective devices. 1.4 Understand importance of proper installation, calibration, and safety compliance. 1.5 Procedures for mounting, connecting, and testing distribution boards and components. 1.6 Identify phases, neutral, earthing systems, and correct connection methods. 1.7 Understand safe operation, load handling, and protection

	requirements. 1.8 Procedures for cable preparation, connection, testing, inspection, and fault rectification, with understanding of quality standards and compliance.
2. Underpinning Skills	2.1 Selecting cables, cable trays, LT/HT meters, and distribution components based on design and load requirements 2.2 Installing cable trays, supports, and routing systems to ensure proper alignment and load handling 2.3 Performing cable laying, dressing, termination, and connection using appropriate tools and methods 2.4 Installing and configuring LT/HT meters and power distribution components in line with standards 2.5 Applying earthing and bonding techniques to ensure electrical safety and system reliability 2.6 Installing distribution boards with correct positioning, connections, and protective devices 2.7 Performing cable termination using lugs, glands, and ferrules with secure and insulated connections 2.8 Carrying out wiring activities following approved methods and circuit layouts 2.9 Performing electrical testing such as continuity and insulation resistance and recording results
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed cables, cable trays, LT/HT meters 1.2 installed power distribution components 1.3 installed distribution boards 1.4 terminated and connected power and control cables 1.5 installed power supply components 1.6 carried out wiring and testing

	1.7 inspected installation quality and alignment with design
2 Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3 Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE AND MAINTAIN ELECTRICAL EQUIPMENT**	Nominal Duration: 80 Hrs.	Unit Code: SICIP-BPI-EWE-05-O

Unit Descriptor:

This unit covers the skills, knowledge and attitudes required to operate and maintain electrical equipment. It specifically includes the tasks of understanding working principles of generator and transformer, operating generator and associated control systems, operating transformer and associated switchgear, understanding battery bank and battery charger, performing routine maintenance of electrical equipment and diagnosing faults and carrying out basic troubleshooting.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand working principles of generator and transformer	1.1 Basic operating principles of generators and transformers are explained. 1.2 **Key components** are identified and their functions described. 1.3 Types of generators and transformers are distinguished based on application, voltage level, and construction. 1.4 Input-output relationships, voltage transformation, and load behavior are interpreted using standard concepts. 1.5 **Safety considerations** and operational limits are recognized to ensure proper use and maintenance.
2. Operate generator and associated control systems	2.1 **Operating procedures and control panel layouts** are interpreted to prepare for generator operation. 2.2 Pre-start checks are conducted. 2.3 Generator is started, synchronized (if required), and brought to load following standard operating procedures. 2.4 **Control systems and instruments** are monitored to maintain stable voltage, frequency, and load conditions. 2.5 Abnormal conditions are identified and shutdown procedures are carried out safely.
3. Operate transformer and associated switchgear	3.1 **Operating procedures, single-line diagrams, and switchgear layouts** are interpreted. 3.2 **Pre-operation checks** are carried out on transformer oil level, temperature indicators and protection systems. 3.3 Transformer is energized and de-energized using approved switching procedures. 3.4 **Switchgear** is operated and monitored to maintain system stability. 3.5 Abnormal conditions are identified and appropriate corrective or shutdown actions are taken.
4. Understand Battery Bank and battery charger.	4.1 Working principles of battery banks and chargers are explained. 4.2 **Types of batteries and chargers** are identified. 4.3 Key components are described with their functions.

	4.4 Charging methods, voltage levels, and safety parameters are interpreted. 4.5 Hazards and ventilation requirements are recognized and managed.
5. Perform routine maintenance of electrical equipment	5.1 Maintenance schedules, manuals, and checklists are interpreted to plan routine maintenance activities. 5.2 Equipment is isolated, de-energized, and secured following lockout/tagout procedures before maintenance. 5.3 Cleaning, tightening, lubrication, and minor adjustments are carried out using appropriate tools and methods. 5.4 Components are inspected for wear, damage, overheating, or corrosion and defects are identified. 5.5 Functional checks and basic tests are performed to ensure equipment operates within specified parameters. 5.6 **<u>Maintenance activities</u>** and findings are recorded, and faults are reported for corrective action.
6. Diagnose faults and carry out basic troubleshooting	6.1 **<u>Fault symptoms</u>** are identified through observation, and operator feedback. 6.2 Wiring diagrams, manuals, and troubleshooting guides are interpreted to trace possible fault sources. 6.3 Basic diagnostic tests and current measurements are performed using appropriate instruments. 6.4 Faults in components, connections, or circuits are isolated and identified systematically. 6.5 Minor faults are rectified using approved methods, and equipment is restored to safe working condition. 6.6 Unresolved or major faults are reported with clear documentation for further technical intervention.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Key components	**Generators:** 1.1 Stator and rotor 1.2 Field windings and armature windings 1.3 Slip rings and brushes 1.4 Prime mover (engine or turbine) 1.5 Voltage regulator **Transformers:** 1.6 Primary and secondary windings 1.7 Magnetic core (laminated core) 1.8 Insulation system 1.9 Oil tank and cooling system (for power transformers) 1.10 Tap changer (where applicable)
2. Safety considerations	2.1 Proper earthing and grounding 2.2 Overload and short-circuit protection 2.3 Insulation integrity and clearance distances 2.4 Safe handling of high voltage equipment

	2.5 Fire and thermal protection (especially for oil-filled transformers)
3. Operating procedures and control panel layouts	3.1 Standard operating procedures (SOPs) for generator operation 3.2 Control panel diagrams and mimic layouts 3.3 Manufacturer manuals and operating instructions 3.4 Start-up, synchronization, loading, and shutdown procedures 3.5 Emergency operating procedures
4. Control systems and instruments	4.1 Voltmeter 4.2 Ammeter 4.3 Frequency meter 4.4 Power factor meter and energy meter 4.5 Control switches 4.6 Indicators 4.7 Alarms 4.8 Automatic Voltage Regulator (AVR) 4.9 Governor control system
5. Operating procedures, single-line diagrams, and switchgear layouts	5.1 Standard operating procedures (SOPs) for transformer and switchgear operation 5.2 Single-line diagrams (SLD) 5.3 Switchgear and substation layout drawings 5.4 Switching schedules and sequence charts 5.5 Manufacturer manuals and operational guidelines
6. Pre-operation checks	6.1 Transformer oil level and oil condition (color, leakage) 6.2 Winding and oil temperature indicators 6.3 Buchholz relay status (for oil-filled transformers) 6.4 Silica gel breather condition 6.5 Protection systems (relays, alarms, trip circuits) 6.6 Earthing connections and physical condition
7. Switchgear	7.1 Circuit breakers (VCB, ACB, SF6 breakers) 7.2 Isolators and disconnect switches 7.3 Load break switches 7.4 Control panels and protection panels 7.5 Busbars and associated equipment
8. Types of batteries and chargers	**Batteries** 8.1 Lead-acid batteries (flooded/wet type) 8.2 Sealed maintenance-free (SMF) batteries 8.3 Valve-regulated lead-acid (VRLA) batteries 8.4 Gel and AGM (Absorbent Glass Mat) batteries 8.5 Lithium-ion batteries (where applicable) **Chargers** 8.6 Constant voltage chargers 8.7 Constant current chargers 8.8 Automatic/smart chargers 8.9 Float chargers and boost chargers 8.10 Trickle chargers
9. Maintenance activities	9.1 Cleaning dust, dirt, and debris from equipment

	9.2 Tightening terminals, bolts, and connections 9.3 Lubrication of moving parts (where applicable) 9.4 Minor adjustments and alignment corrections 9.5 Replacement of small consumables (fuses, connectors)
10. Fault symptoms	10.1 Equipment not starting or operating intermittently 10.2 Tripping of circuit breakers or protection devices 10.3 Abnormal noise, vibration, or smell 10.4 Overheating of components or cables 10.5 Voltage drops or fluctuation 10.6 Indicator alarms or warning signals 10.7 Feedback from operators or users

Curricular Content Guide

1. Underpinning Knowledge	1.1. Procedures for generator operation, including pre-start checks, synchronization, and shutdown. 1.2. Identify types of generators, transformers, and their key components. 1.3. Understand voltage transformation, load behavior, and safety limits. 1.4. Procedures for energizing, de-energizing, and monitoring transformers and switchgear. 1.5. Understand importance of safe operation and system stability. 1.6. Procedures for charging, monitoring, and maintaining batteries. 1.7. Identify battery types, chargers, and components. 1.8. Understand safety hazards and ventilation requirements. 1.9. Procedures for inspection, testing, fault diagnosis, and rectification. 1.10. Identify faults, defects, and affected components.
2. Underpinning Skills	2.1 Applying knowledge of generator and transformer working principles, components, and operating limits 2.2 Operating generators by following standard procedures, performing pre-start checks, and monitoring control systems 2.3 Maintaining stable voltage, frequency, and load conditions during generator operation 2.4 Monitoring transformer conditions and identifying abnormal situations for safe shutdown 2.5 Applying safety measures related to battery systems, including hazard control and ventilation requirements 2.6 Performing routine maintenance activities such as cleaning, tightening, inspection, and minor adjustments 2.7 Using lockout/tagout procedures to ensure safe isolation before maintenance work 2.8 Diagnosing faults by interpreting diagrams, conducting tests, and identifying issues systematically 2.9 Performing basic troubleshooting and reporting major faults

	with proper documentation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 understood working principles of generator and transformer 1.2 operated generator and associated control systems 1.3 operated transformer and associated switchgear 1.4 understood Battery Bank and battery charger. 1.5 performed routine maintenance of electrical equipment 1.6 diagnosed faults and carried out basic troubleshooting
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INSTALL EARTHING SYSTEM**	**Nominal Duration:** 30 Hrs.	**Unit Code:** SICIP-BPI-EWE-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required install earthing system. It specifically includes the tasks of interpreting earthing system requirements, preparing tools, equipment and materials, preparing site for earthing installation, installing earthing electrode and components, testing earthing system performance and completing installation and documentation.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret earthing system requirements	1.1 Earthing system drawings, specifications, and standards are interpreted to determine installation requirements. 1.2 **Types of earthing systems** are identified based on-site conditions and application. 1.3 Soil resistivity, load characteristics, and fault current requirements are analyzed. 1.4 Materials, conductor sizes, and **layout configurations** are determined in line with safety and regulatory standards. 1.5 Safety requirements and compliance criteria are clearly understood before installation.
2. Prepare tools, equipment, and materials	2.1 Tools are selected as per job requirements. 2.2 Tools and equipment are inspected for serviceability, calibration, and safety compliance before use. 2.3 Required quantities of materials are verified and prepared according to the installation plan. 2.4 Work area is organized and all resources are arranged to ensure safe and efficient installation.
3. Prepare site for earthing installation	3.1 Site layout and earthing design are reviewed to determine electrode locations and installation points. 3.2 Area is cleared, marked, and made safe for excavation and installation activities. 3.3 **Excavation or drilling** is carried out to required depth and dimensions as per specifications. 3.4 Soil condition is assessed and treated (if required) to achieve desired earth resistance levels. 3.5 Safety measures are implemented to prevent hazards during site preparation and installation.
4. Install earthing electrode and components	4.1 **Earthing electrodes** are installed at designated locations as per design specifications. 4.2 Earthing conductors are connected to electrodes using approved clamps, welding, or brazing methods. 4.3 **Backfilling** is carried out using suitable materials to improve earth conductivity.

	4.4 Earthing pits, inspection chambers, and covers are constructed and secured for accessibility and protection. 4.5 All components are aligned, fixed, and protected against mechanical damage and corrosion.
5. Test earthing system performance	5.1 Testing procedures and standards are interpreted to determine required earth resistance and test methods. 5.2 **<u>Earth resistance</u>** is measured using appropriate instruments following standard testing techniques. 5.3 Continuity of earthing conductors and connections is verified to ensure effective grounding. 5.4 Test results are compared with permissible limits to confirm compliance with safety standards. 5.5 **<u>Deficiencies</u>** are identified and corrective actions are taken to achieve required performance.
6. Complete installation and documentation	6.1 Installed earthing system is inspected to ensure all components are properly connected, secured, and protected. 6.2 Site is cleaned, restored, and made safe in accordance with workplace procedures. 6.3 **<u>Final checks</u>** are conducted to confirm compliance with design specifications and safety standards. 6.4 Installation details, test results, and materials used are recorded accurately in required formats. 6.5 Documentation is submitted and handed over as per organizational and regulatory requirements.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Types of earthing systems	1.1 Plate earthing (copper or GI plate) 1.2 Rod earthing (driven electrode) 1.3 Strip or wire earthing 1.4 Grid or mesh earthing (substation grounding) 1.5 Chemical earthing systems 1.6 Equipment earthing and system (neutral) earthing
2. Layout configurations	2.1 Single electrode or multiple electrode systems 2.2 Ring earthing around buildings 2.3 Grid or mesh earthing for substations 2.4 Bonding of all metallic parts and structures 2.5 Separation or interconnection of earthing systems
3. Excavation or drilling	3.1 Manual digging using spades, shovels 3.2 Mechanical excavation (auger drilling, machine digging) 3.3 Drilling for rod or pipe electrodes 3.4 Excavation of pits or trenches for plate or strip earthing 3.5 Maintaining specified depth, width, and spacing
4. Earthing electrodes	4.1 Rod electrodes (copper, copper-bonded, GI rods) 4.2 Plate electrodes (copper plate, GI plate) 4.3 Pipe electrodes (perforated GI pipes)

	4.4 Chemical/maintenance-free earthing electrodes
5. Backfilling	5.1 Place electrode or conductor in the pit/trench 5.2 Add layers of selected backfill material 5.3 Compact each layer properly to remove air gaps 5.4 Maintain moisture (if required for conductivity) 5.5 Ensure final surface is level and stable
6. Earth resistance	6.1 Fall-of-potential (three-point test) 6.2 Clamp-on earth resistance testing (without disconnection) 6.3 Two-point method (for simple systems) 6.4 Soil resistivity testing (supporting measurement)
7. Deficiencies	7.1 High earth resistance values 7.2 Poor or loose connections 7.3 Corrosion or damage of conductors/electrodes 7.4 Inadequate soil conductivity 7.5 Improper installation or spacing of electrodes
8. Final checks	8.1 Confirmation of installation as per approved drawings 8.2 Verification of earthing continuity and connections 8.3 Review of earth resistance test results 8.4 Compliance with design specifications and tolerances 8.5 Functional readiness of the earthing system

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify soil conditions, materials, conductor sizes, and layout configurations. 1.2 Procedures for selecting, inspecting, and organizing tools and resources. 1.3 Understand importance of readiness for safe and efficient work. 1.4 Concepts of site preparation and installation planning. 1.5 Identify installation locations and site conditions. 1.6 Understand safety measures during site work. 1.7 Identify clamps, joints, and protective components. 1.8 Understand need for durability and corrosion protection. 1.9 Identify testing instruments and acceptable limits. 1.10 Understand importance of compliance with safety standards. 1.11 Identify defects, records, and documentation requirements. 1.12 Understand importance of proper handover and compliance.
2. Underpinning Skills	2.1 Analyzing soil resistivity, load conditions, and fault current to select suitable earthing systems 2.2 Selecting tools, equipment, and materials based on job requirements and safety standards 2.3 Preparing and organizing the work area to ensure safe and efficient installation 2.4 Planning site layout, marking locations, and performing excavation as per design specifications 2.5 Applying soil treatment methods to achieve required earth

	resistance levels 2.6 Installing earthing electrodes, conductors, and components using approved methods 2.7 Performing backfilling and constructing earthing pits with proper protection and accessibility 2.8 Conducting earth resistance and continuity tests using appropriate instruments 2.9 Evaluating test results and taking corrective actions to meet safety standards 2.10 Inspecting completed installation for alignment, protection, and compliance
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted earthing system requirements 1.2 prepared tools, equipment, and materials 1.3 prepared site for earthing installation 1.4 installed earthing electrode and components 1.5 tested earthing system performance 1.6 completed installation and documentation
2 Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3 Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM MOTOR CONNECTION**	Nominal Duration: 40 Hrs.	Unit Code: SICIP-BPI-EWE-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform motor connection. It specifically includes the tasks of identifying and selecting control and protective devices, collecting tools, equipment and materials, installing control and protective devices, performing motor connection, checking and testing circuit and cleaning the workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and select control and protective devices	1.1 **Manuals** and documents of controlling and protective devices are collected. 1.2 **Drawings** and **symbols** of controlling and protective devices are sorted. 1.3 Types of **controlling** and **protective devices** are listed.
2. Collect tools, equipment and materials	2.1 Tools, equipment and materials are identified and collected. 2.2 Tools, Equipment and Materials are checked for usability, 2.3 PPE is collected and used.
3. Install control and protective devices	3.1 Control and **protective devices** are selected and collected according to the need of the operations. 3.2 Control and protective devices are installed according to the layout plan. 3.3 Control and protective devices are set and connected to the motor.
4. Perform motor connection	4.1 Direct on-line starter is collected and its diagram is interpreted. 4.2 Direct on-line starter is connected with the motor. 4.3 Star-delta starter is collected and its diagram interpreted. 4.4 Star-delta starter is connected with the motor. 4.5 Reverse-Forward starter is collected and its diagram interpreted. 4.6 Reverse-Forward starter is connected with the motor.
5. Check and test circuit	5.1 All the connections of each starter are checked and justified. 5.2 Connection between motor and starter is checked and tested.
6. Clean the workplace	6.1 Cleaning tools and equipment are selected & collected. 6.2 Cleaning tools and equipment are prepared for cleaning. 6.3 Waste materials are disposed. 6.4 Cleaning is completed.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Manuals	1.1 Manufacturer's specification manual 1.2 Repair manual 1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Manual of instruction
2. Drawings	2.1 Technical drawing 2.2 Sketch 2.3 Blue print 2.4 Electrical drawings 2.5 Connection diagram
3. Symbols	3.1 Drawing symbol 3.2 Connection symbol 3.3 Load symbol 3.4 Socket symbol 3.5 Main switch symbol 3.6 Supply symbol 3.7 Danger symbol 3.8 Switch board symbol 3.9 Conduit symbol 3.10 Starter symbol 3.11 Protective device symbol 3.12 Motor symbol
4. Controlling Devices	4.1 Push Switches 4.2 Main switches 4.3 Direct on-line starter 4.4 Change over switches 4.5 Timer 4.6 Relay 4.7 Magnetic contactor
5. Protective devices	5.1 Circuit breaker 5.2 Over load Relays 5.3 MCCB 5.4 ELCB

Curricular Content Guide

1. Underpinning Knowledge	1.1. Procedures for interpreting manuals, drawings, and symbols. 1.2. Understand correct selection based on application and safety. 1.3. Procedures for collecting and checking tools and materials. 1.4. Understand importance of usability and safety

	compliance. 1.5. Identify layout requirements and connection points. 1.6. Understand importance of correct installation for safe operation. 1.7. Procedures for connecting DOL, star-delta, and reverse-forward starters. 1.8. Procedures for inspecting and testing connections. 1.9. Understand importance of proper testing for safe operation. 1.10. Procedures for cleaning tools, equipment, and work area. 1.11. Identify waste materials and disposal methods.
2. Underpinning Skills	2.1. Identifying and selecting appropriate control and protective devices based on operational needs 2.2. Installing control and protective devices according to layout plans and specifications 2.3. Performing correct setting and connection of devices with motors 2.4. Performing motor connections using standard procedures and wiring methods 2.5. Checking and verifying all electrical connections for accuracy and safety 2.6. Testing motor and starter circuits to ensure proper operation 2.7. Applying safe and organized workplace practices during and after work 2.8. Cleaning work area, handling waste materials, and maintaining tools properly
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected control and protective devices

	1.2 collected tools, equipment and materials 1.3 installed control and protective devices 1.4 performed motor connection 1.5 checked and tested circuit 1.6 cleaned the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Electrical Works in Energy Sector**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training Facilities	Required Quantity
1.	Computers/Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	Generators (25 KVA) Diesel operated	01
5.	Transformers (3-5KVA) single phase 220/110V	03
6.	Circuit breakers (SPMCB, DPMCB, TPMCB)	06
7.	Switchgear (CB, LBS, Relays, Fuses, CT, PT)	03 (Each)
8.	3-phase induction Motor (3 to 5 hp.)	03
9.	Motor Control panels (Star-Delta)	05 sets
10.	Electrical goods/fittings	10 for each
11.	Battery (50 AH, 12V) with charger	03
12.	Distribution boards	05
13.	Earthing equipment	03
14.	Lightning protection system	03
15.	Cable/Wire (1rm, 1.5rm, 2.5rm & 1 coil flexible wire)	10 coils
16.	Fire extinguisher	04
17.	First aid box	03
Major Tools		**Required Quantity**
18.	Screwdrivers (insulated)	10
19.	Combination pliers	10
20.	Wire stripper and cutter	10
21.	Crimping pliers	10
22.	Tester	10
23.	Voltage tester	10
24.	Cable pulling tools	10
25.	Drilling machine	10
26.	Hack Saw	10
27.	Pipe cutter	10
28.	Multimeter	10
29.	Clamp meter	10
30.	Watt meter	10

31.	Insulation resistance tester (Megger)	10
32.	Earth resistance tester	10
33.	Phase sequence tester	10
34.	Continuity tester	10
35.	OHM tester	10
36.	Voltage detector	10
37.	Power quality analyzer	10
38.	Loop impedance tester	10
39.	Tool box set	10

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Electrical Works in Energy Sector** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.